
```
%EA_HW_3_1&2 Nouh Shaikh
clc; clear; close all;
```

Problem 1:

```
% 1:
A = [ 2,  3,  1, -1;
      5, -2,  5, -4;
      1, -2,  3, -3;
      3, -8, -1,  1];
b = [1; 5; 3; -1];

fprintf('--- Step 1: Matrix A and Vector b ---\n');
disp('A ='); disp(A);
disp('b ='); disp(b);

% 2:
Aug = [A, b];
fprintf('\n--- Step 2: Augmented Matrix [A|b] ---\n');
disp('Aug ='); disp(Aug);

% 3:
n = size(A, 1);
L = eye(n);
U_aug = Aug;

for j = 1:n-1
    for i = j+1:n
        multiplier = U_aug(i,j) / U_aug(j,j);
        L(i,j) = multiplier;
        U_aug(i, j:end) = U_aug(i, j:end) - multiplier * U_aug(j, j:end);
    end
end

U = U_aug(:, 1:n);
b_prime = U_aug(:, end);

fprintf('\n--- Step 3: Upper Triangular Matrix U and Multipliers ---\n');
disp('U ='); disp(U);
disp('L (Multipliers matrix) ='); disp(L);

% 4:
x_back = zeros(n, 1);
for i = n:-1:1
    x_back(i) = (b_prime(i) - U(i, i+1:end) * x_back(i+1:end)) / U(i,i);
end
fprintf('\n--- Step 4: Back Substitution Solution ---\n');
disp('x ='); disp(x_back);

% 5:
A_check = L * U;
fprintf('\n--- Step 5: Verification (A = L * U) ---\n');
```

```

disp('L * U ='); disp(A_check);

% 6:
d = zeros(n, 1);
for i = 1:n
    d(i) = (b(i) - L(i, 1:i-1) * d(1:i-1)) / L(i,i);
end

x_lu = zeros(n, 1);
for i = n:-1:1
    x_lu(i) = (d(i) - U(i, i+1:end) * x_lu(i+1:end)) / U(i,i);
end

fprintf('\n--- Step 6: LU Factorization Solution ---\n');
disp('d ='); disp(d);
disp('x (via LU) ='); disp(x_lu);

% 7:
x_div = A \ b;
fprintf('\n--- Step 7: MATLAB Left Division Solution ---\n');
disp('x (via A \ b) ='); disp(x_div);

--- Step 1: Matrix A and Vector b ---
A =
    2     3     1    -1
    5    -2     5    -4
    1    -2     3    -3
    3    -8    -1     1

b =
    1
    5
    3
   -1

--- Step 2: Augmented Matrix [A|b] ---
Aug =
    2     3     1    -1     1
    5    -2     5    -4     5
    1    -2     3    -3     3
    3    -8    -1     1    -1

--- Step 3: Upper Triangular Matrix U and Multipliers ---
U =
    2.0000    3.0000    1.0000   -1.0000
         0   -9.5000    2.5000   -1.5000
         0         0    1.5789   -1.9474
         0         0         0   -2.6667

L (Multipliers matrix) =
    1.0000         0         0         0
    2.5000    1.0000         0         0

```

0.5000	0.3684	1.0000	0
1.5000	1.3158	-3.6667	1.0000

--- Step 4: Back Substitution Solution ---

x =

0
0
1
0

--- Step 5: Verification (A = L * U) ---

L * U =

2	3	1	-1
5	-2	5	-4
1	-2	3	-3
3	-8	-1	1

--- Step 6: LU Factorization Solution ---

d =

1.0000
2.5000
1.5789
0

x (via LU) =

0
0
1
0

--- Step 7: MATLAB Left Division Solution ---

x (via A \ b) =

0
0
1
0

Problem #2:

```
fprintf('--- Problem #2: Matrix Inverse via LU ---\n');
```

```
I_mat = eye(n);
```

```
A_inv = zeros(n, n);
```

```
for col = 1:n
```

```
    e_i = I_mat(:, col);
```

```
    d_i = zeros(n, 1);
```

```

    for i = 1:n
        d_i(i) = (e_i(i) - L(i, 1:i-1) * d_i(1:i-1)) / L(i,i);
    end

    x_i = zeros(n, 1);
    for i = n:-1:1
        x_i(i) = (d_i(i) - U(i, i+1:end) * x_i(i+1:end)) / U(i,i);
    end

    A_inv(:, col) = x_i;
end

fprintf('\nComputed Inverse Matrix A^-1:\n');
disp(A_inv);

fprintf('MATLAB Native inv(A) for comparison:\n');
disp(inv(A));

x_from_inv = A_inv * b;

fprintf('\nVerification: Solution vector {x} via A^-1 * b:\n');
disp(x_from_inv);

fprintf('Original solution vector {x} from Problem 1:\n');
disp(x_back);

difference = norm(x_from_inv - x_back);
fprintf('Absolute numerical difference between the two methods: %e\n',
difference);

--- Problem #2: Matrix Inverse via LU ---

Computed Inverse Matrix A^-1:
    0.3250         0   -0.0625    0.1375
    0.1250         0   -0.0625   -0.0625
   -1.2750    1.0000   -1.0625   -0.4625
   -1.2500    1.0000   -1.3750   -0.3750

MATLAB Native inv(A) for comparison:
    0.3250   -0.0000   -0.0625    0.1375
    0.1250   -0.0000   -0.0625   -0.0625
   -1.2750    1.0000   -1.0625   -0.4625
   -1.2500    1.0000   -1.3750   -0.3750

Verification: Solution vector {x} via A^-1 * b:
         0
   -0.0000
    1.0000
   -0.0000

Original solution vector {x} from Problem 1:
         0
         0

```

$\frac{1}{0}$

Absolute numerical difference between the two methods: 5.721958e-17

Published with MATLAB® R2026a